

STAGE 01 · PROBLEM STAGE 02 · PIPELINE STAGE 03 · DISCOVERY STAGE 04 · EVALUATION

STAGE 05 · DECISION TOOL

OIYAS AI HACKATHON · CROP YIELD REGRESSION

# Predicting Crop Yield Before Planting Decisions

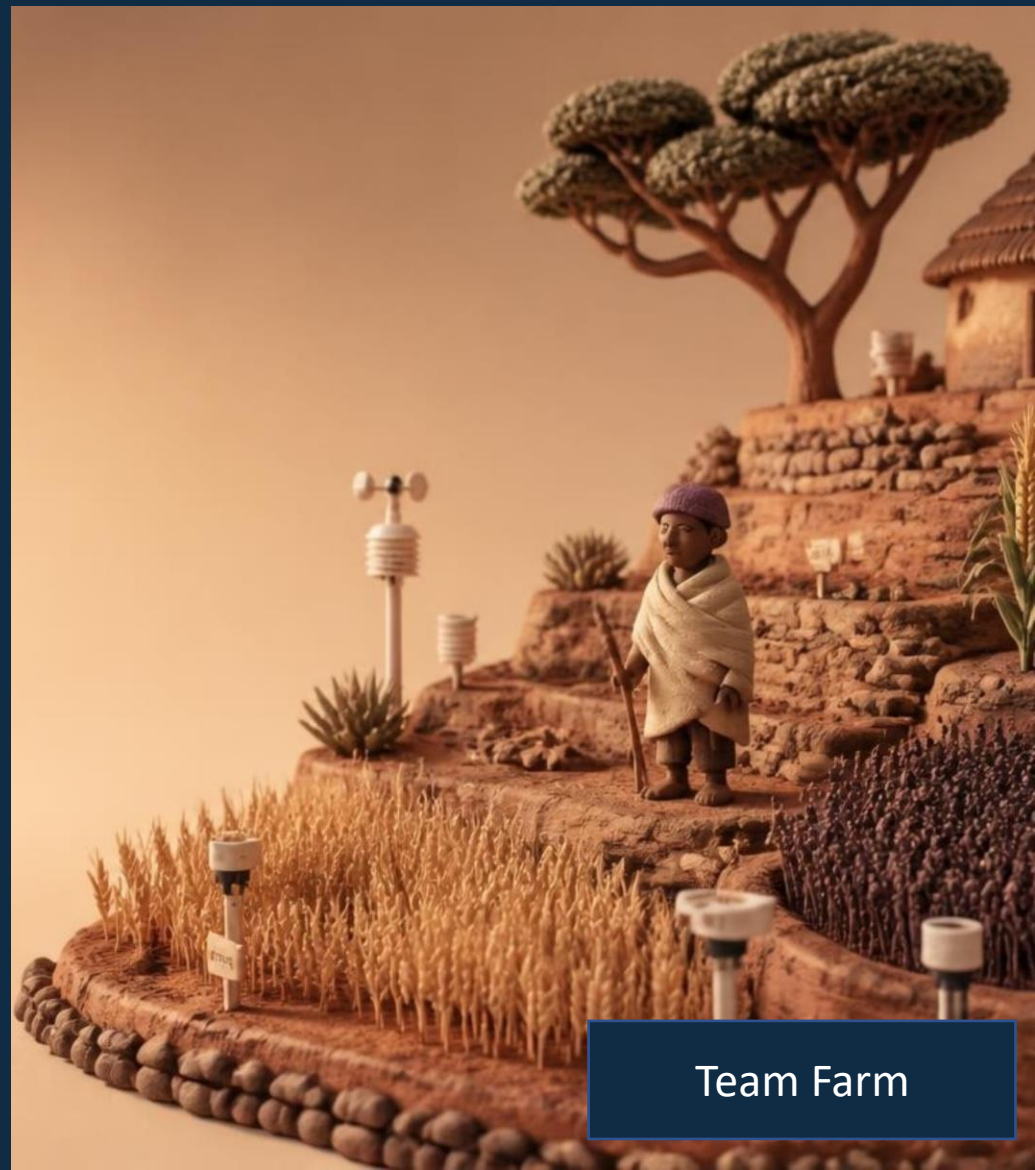
Ethiopian smallholder regression · five regions · five crops

*Can we predict crop yield before planting using only information available at or before planting time?*

**Inputs** · plot survey data · regional weather · market prices

**TARGET** : YIELD\_TONS\_PER\_HA

FOR EACH LEADERBOARD PLOT, USING ONLY INFORMATION KNOWN AT OR BEFORE PLANTING TIME.



Team Farm

# QIYAS HACKATHON

# Team Farm



## TEAM MEMBERS :

1. Desalegn Mengsitu
2. Abreham Cheru
3. Regina Giovanni
4. Henok Solomon
5. Mohammed Omer

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SUNMITED ON

October 6, 2026





# Data sources

## PLOT SURVEY

15,090 plots · 2021–2024 · region · crop · planting information · plot characteristics

## REGIONAL WEATHER

5 regions · 4 years · 12 months/year · rainfall · temperature · humidity

## MARKET PRICES

5 crops · 5 regions · 4 years · market price · used for revenue estimation

# Three exports → one trustworthy modeling table

## Data-quality problems

**REGION LABELS:** 4+ casing/label variants and abbreviations such as AMH/SOM

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**CROP LABELS:** Tef / teff spelling mismatch

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**MISSING VALUES:** -999 and -999.0 sentinels

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**PRICE UNITS:** Birr/kg vs Birr/quintal

## Join logic

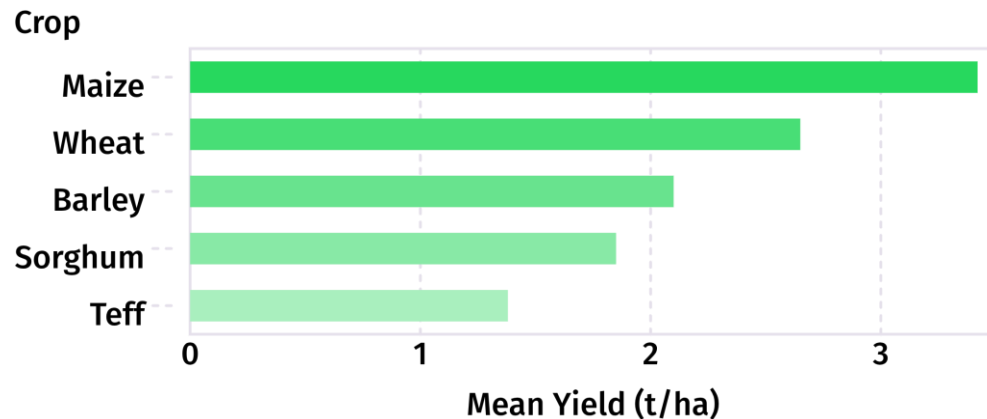
PLOT TABLE · LEFT JOIN ↓ SEASONAL WEATHER ↓  
PRICE DATA ↓ CLEAN MASTER

Monthly regional weather becomes one season-level row per plot. Growing-season rule: planting month + following 3 months, including year wraparound.

# The data reveals where yield and value move together

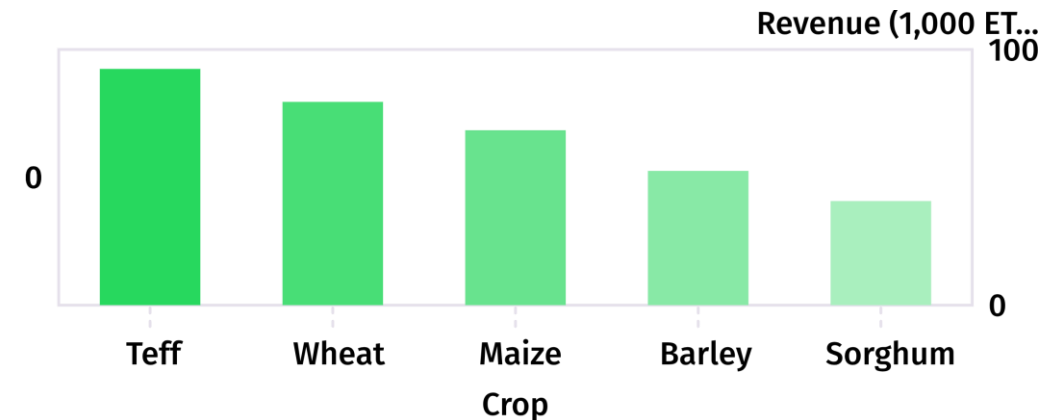
High yield does not always maximize gross value; prices and regional risk reshape smallholder margins.

FIGURE 1 · MEAN YIELD BY CROP (TONS / HA)



**TAKEAWAY:** MAIZE HAS THE HIGHEST AVERAGE YIELD AT 3.90 T/HA, WHILE TEFF IS LOWEST AT 2.00 T/HA — A 1.90 T/HA GAP ACROSS CROPS.

FIGURE 2 · ESTIMATED REVENUE BY CROP (1K ETB / HA)



**TAKEAWAY:** : TEFF HAS THE LOWEST AVERAGE YIELD AT 2.00 T/HA, YET GENERATES THE HIGHEST ESTIMATED GROSS REVENUE AT ≈152K ETB/HA — SHOWING THAT YIELD ALONE DOES NOT DETERMINE ECONOMIC VALUE.

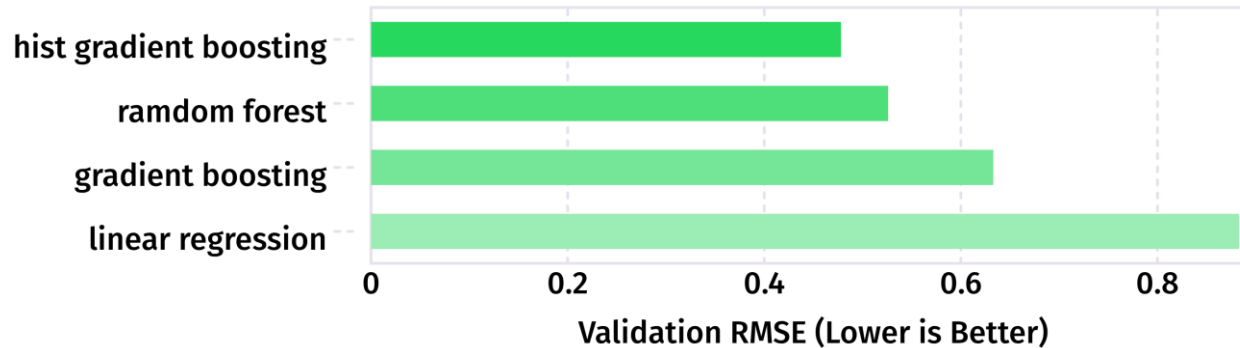
**Agronomic vs economic reality:** Revenue / ha = yield × 10 quintals/ton × price / quintal. Teff has the lowest physical yield per hectare but, at market prices, leads total per-hectare gross revenue.

# Weather features must prove value on held-out evaluation

Leak-free 5-fold cross-validation .

## 5-FOLD CV VALIDATION RMSE · TONS / HA

Model Architect...



5-FOLD CV SPLIT [INSERT SPLIT] · MEAN RMSE 0.480 ± STD ONLY 0.012 — A STABLE RESULT

## KEY TAKEAWAY

Pre-season cumulative rainfall and early moisture indices meaningfully reduced variance in rainfed highland plots.

## WEATHER ABLATION & FINAL METRICS

### SELECTED CHAMPION MODEL

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Hist\_gradient\_boosting

RMSE =0.4778 R<sup>2</sup>= 0.8856



# From prediction error to a working decision tool

Translating regression outputs into an accessible planting advisory application for extension workers.



## LEADERBOARD SCORE

# 0.4778

Final test set RMSE on held-out plots.

## ERROR RESIDUAL ANALYSIS · HARDEST SUBGROUPS

Hardest crop to predict: maize (mean abs. error 0.47 tons/ha).

Easiest: teff (0.27).

Hardest region: tigray (mean abs. error 0.40 tons/ha).  
Easiest: somali (0.20, and zero of the 10 worst-predicted plots were in Somali) — a tigray-specific feature is the top candidate if given more time.

## LIVE DEPLOYED ADVISORY APPLICATION

**Demo :** INSERT

**User inputs:** Region, plot size, anticipated planting month, and selected crop.

**Model outputs:** Expected yield (t/ha), confidence interval, estimated gross revenue (ETB), and historical weather context.

**Next technical step:** [INSERT ONE IMPROVEMENT] ·  
**Operational limitation:** [INSERT ONE LIMITATION]

THE END

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QIYAS HACKATHON

**Thank you**

